Roma Tre

Information Visualization

Simple visualization strategies

M. Patrignani

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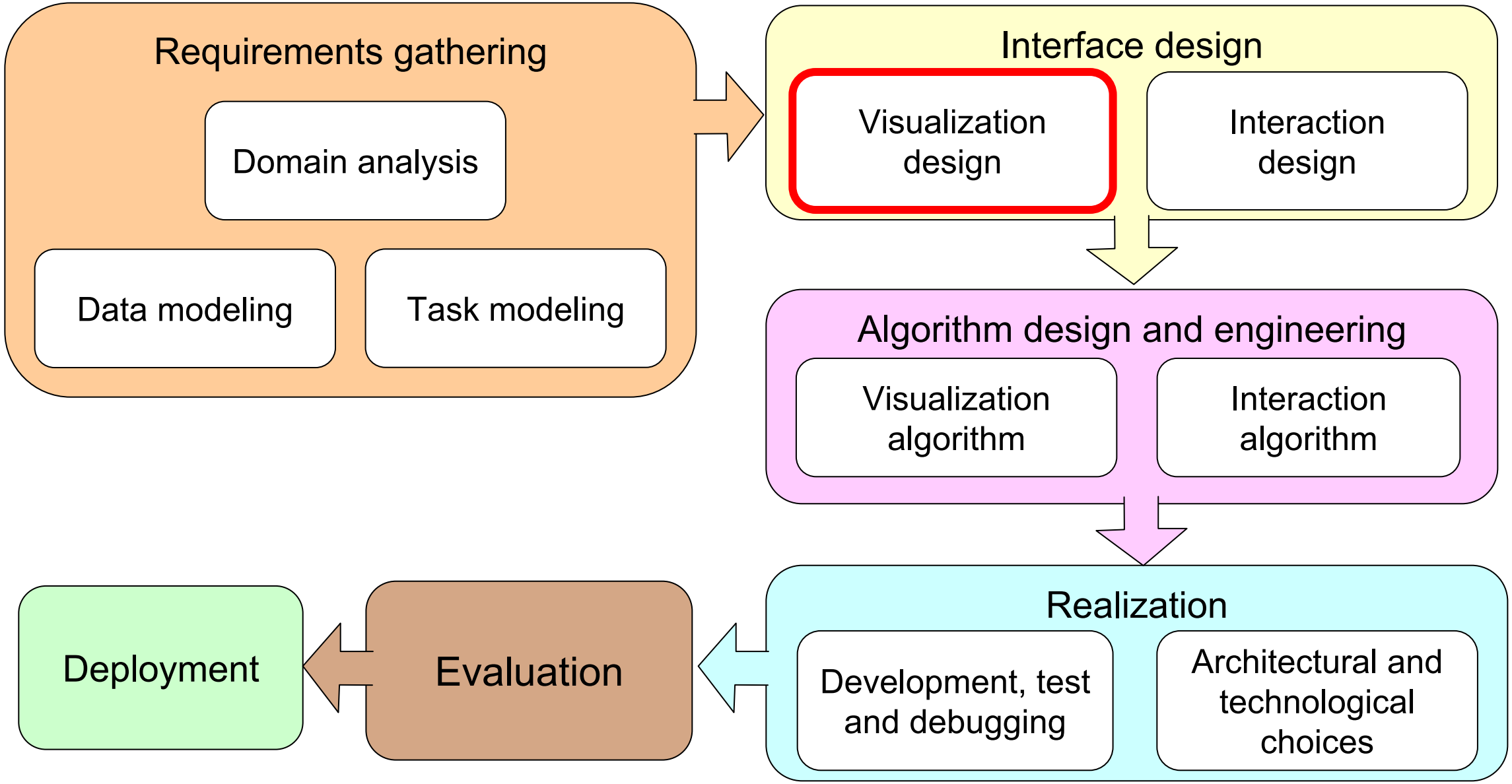
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Overview

- Types of datasets
- Simple flat datasets
  - representations for univariate, bivariate, and trivariate data
- Spacial datasets
  - Thematic maps and cartograms

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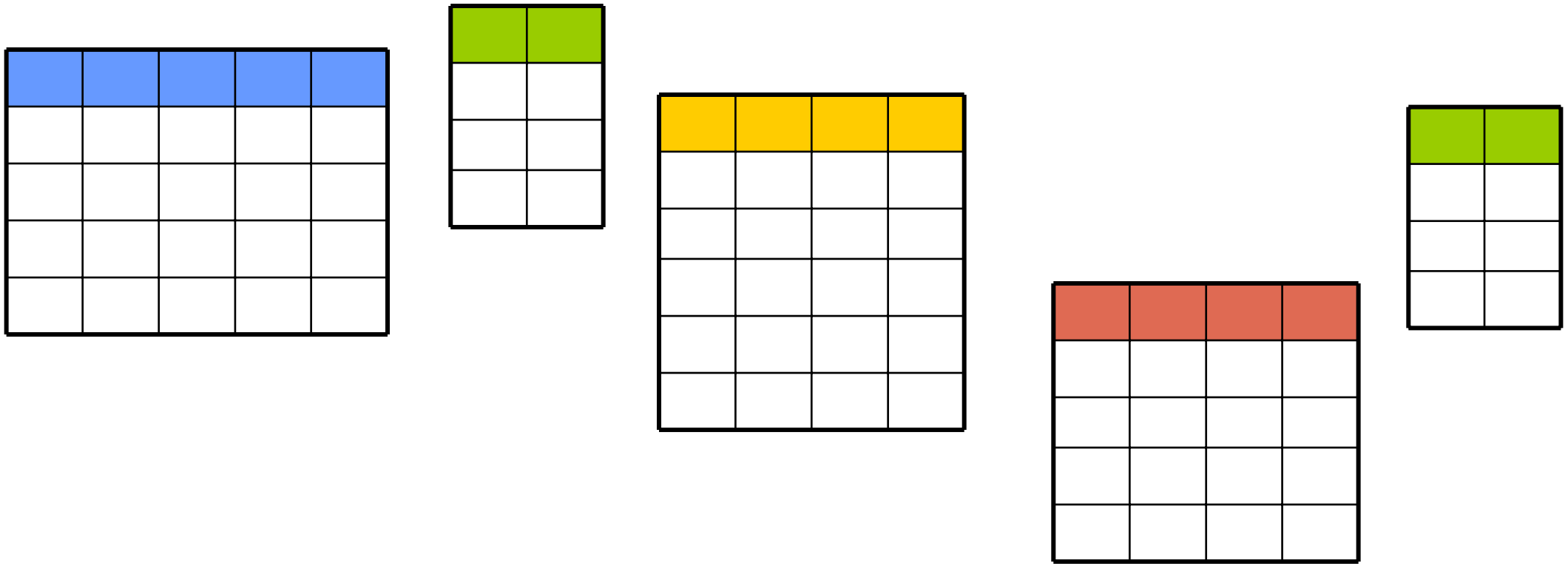
Design pipeline



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Data modeling

- At the end of the data modeling phase we have a collection of data tables



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Dataset types

- Based on the data tables we have in our dataset we can classify the kind of dataset itself
  - flat dataset
  - relational dataset
  - time series
  - geolocated/geospatial data
  - grid/spatial fields
  - …

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Flat datasets visualization

univariate, bivariate and trivariate datasets

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Flat dataset

- The dataset consists of a single table
- This kind of dataset is further classified
  - based on the number of variables
    - univariate, bivariate, trivariate, multivariate
  - based on the kind of variables
    - quantitative, nominal, ordinal
  - based on the independent or dependent variables

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Table representation

- A spreadsheet already does that
  - each variable has its own column
  - each case has its row
- This is actually a mapping

V_1	V_2	V_3	V_4	V_5	V_6
Mary	23L34	A	3	5	3
Luiss	RC576	A	5	1	2
Eric	4G56H	B	3	7	5
Lucy	AP361	A	4	3	4
Frank	2439B	B	6	3	4

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Number of variables

- We can classify or visualization needs by the number of variables (after possible projections)
  - univariate
    - a single variable for each case
  - bivariate
    - two variables
  - trivariate
    - three variables
  - multivariate
    - more than three variables per data case
    - also called “multidimensional” or “hipervariate”

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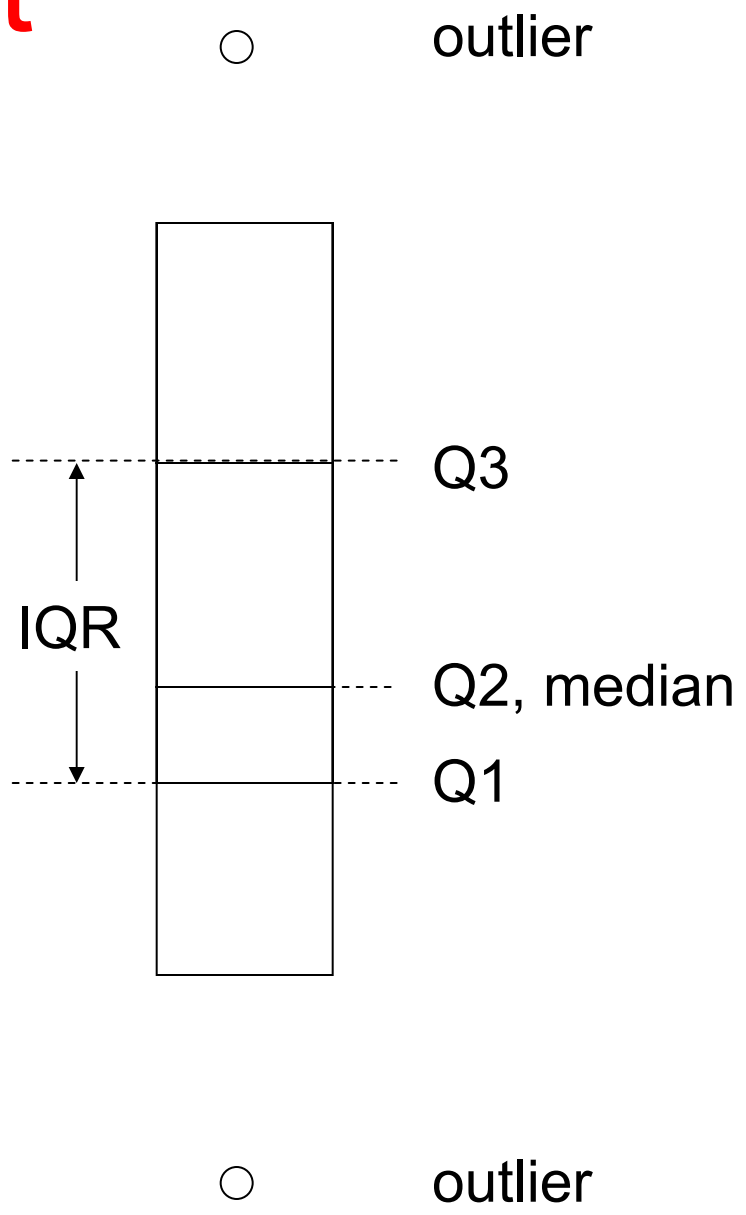
Univariate dataset

- Meaningful features
  - distribution
    - the way data values are spread or arranged, showing frequencies or probabilities of different values (e.g., symmetric, skewed, uniform)
  - average (mean)
    - the sum of all values divided by the number of values
    - represents the “center” of the data but is sensitive to extreme values
  - median
    - the middle value when the data are ordered
      - if there’s an even number of observations, it’s the average of the two middle values
    - robust to extreme values

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Univariate dataset

- Other meaningful features
  - quartiles
    - values that divide ordered data into four equal parts:
      - Q1 (first quartile, 25% of data lie below it)
      - Q2 (second quartile, same as the median)
      - Q3 (third quartile, 75% of data lie below it)
  - outliers:
    - data points that lie unusually far from the rest of the data
    - often defined as Q1−1.5×IQR and Q3+1.5×IQR, where IQR (interquartile range) = Q3−Q1



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Univariate data: stemplots

- Stemplots
  - also called “stem and leaf plot”
  - a single quantitative variable
  - pros
    - succint representation
    - can be easily printed on a text report
    - all values are readable
    - provides a distribution of the input data
  - cons
    - bucket size?
    - encoded, not intuitive

Data: 9, 8, 7, 6, 6, 5, 6, 11, 19, 16, 13, 10, 26, 29, 27, 48, 47, 41, 59, 52, 59, 58, 58, 54, 64, 63, 63, 71, 79, 72, 79, 74, 83, 83, 83, 86, 81, 81, 80, 87

Stems	Leaves
0	9 8 7 6 6 5 6
10	1 9 6 3 0
20	6 9 7
30	
40	8 7 1
50	9 2 9 8 8 4
60	4 3 3
70	1 9 2 9 4
80	3 3 3 6 1 1 0 7

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Stemplots: a bigger example

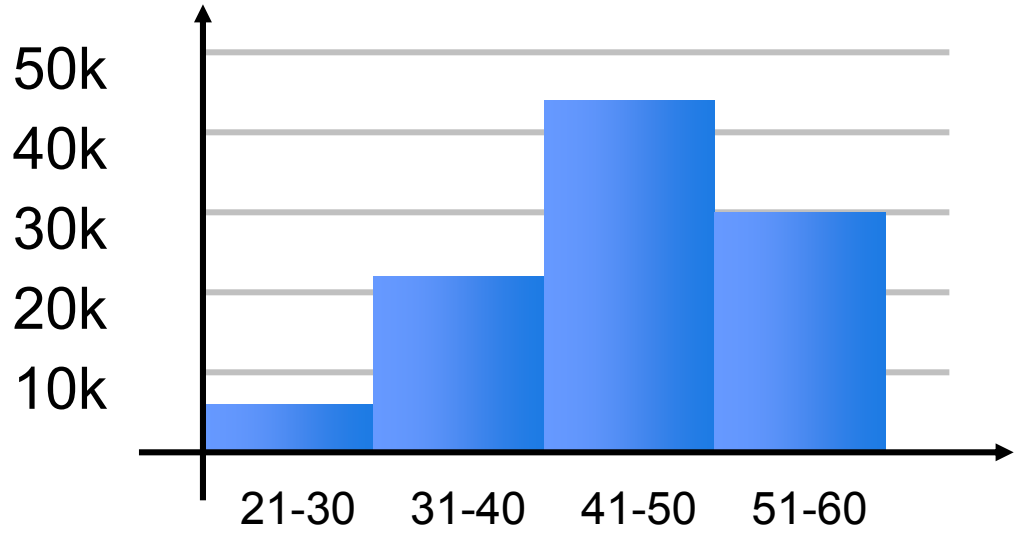
- Stem and leaf plot of the height of 218 volcanoes
  - example from [Turfe 83]

Thousands feet	Hundreds feet
0	98766562
1	97719630
2	69987766544422211009850
3	8766554120995514 26
4	99988443319294333611 07
5	97666666554422210097731
6	898665441077761065
7	98855431100652108073
8	653322122937
9	377655421000493
10	0984433165212
11	4963201631
12	45421164
13	47830
14	00
15	676
16	52
17	92
18	5
19	39730

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Univariate data: histograms

- Histograms
  - one quantitative variable
    - the range of the variable is partitioned into intervals of the same size
    - the diagram shows the number of data points falling in each interval

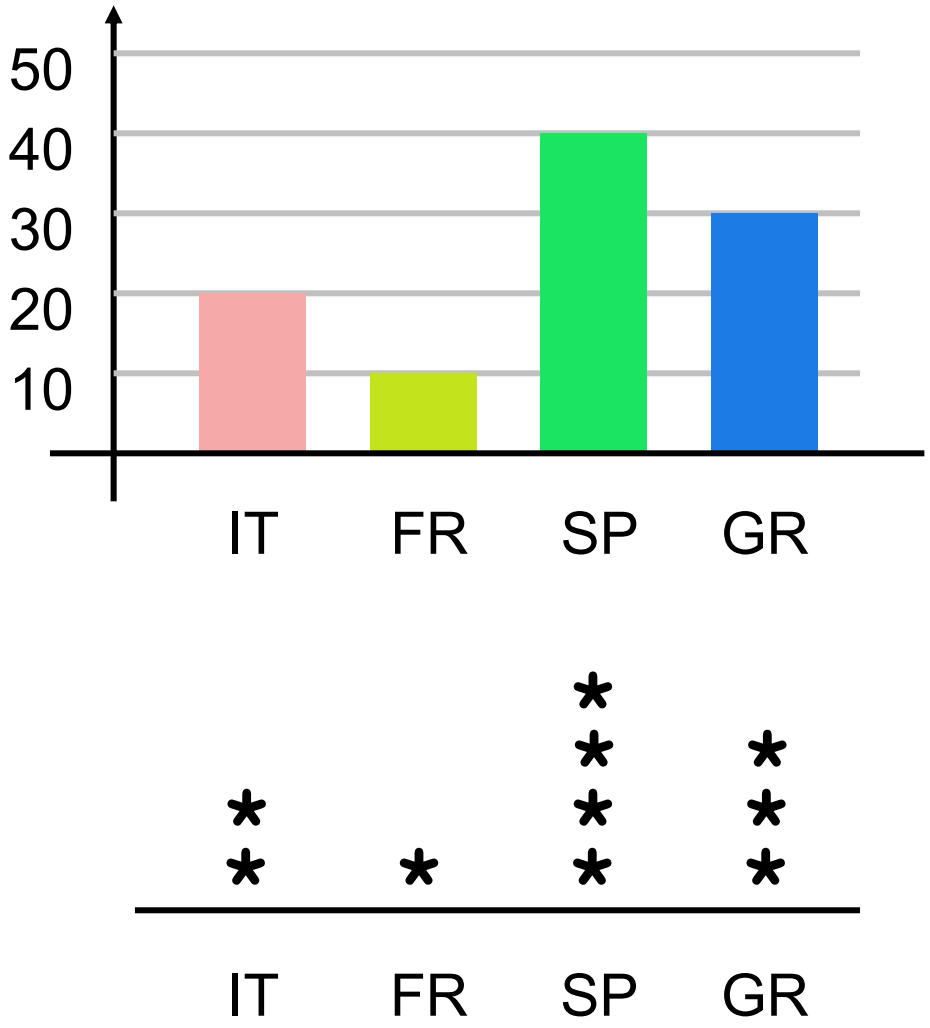


- Histograms represent distributions
  - benefit from position and length

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Bivariate data: bar charts

- Bar charts
  - one categorical variable
    - usually the independent variable
    - mapped to position
      - sometimes also to hue, texture, density, saturation
  - one quantitative variable
    - usually the dependent variable
    - usually continuous
    - mapped both to position and to length
- Dotplots
  - simplified variant used for typewriters
    - discrete quantitative variable



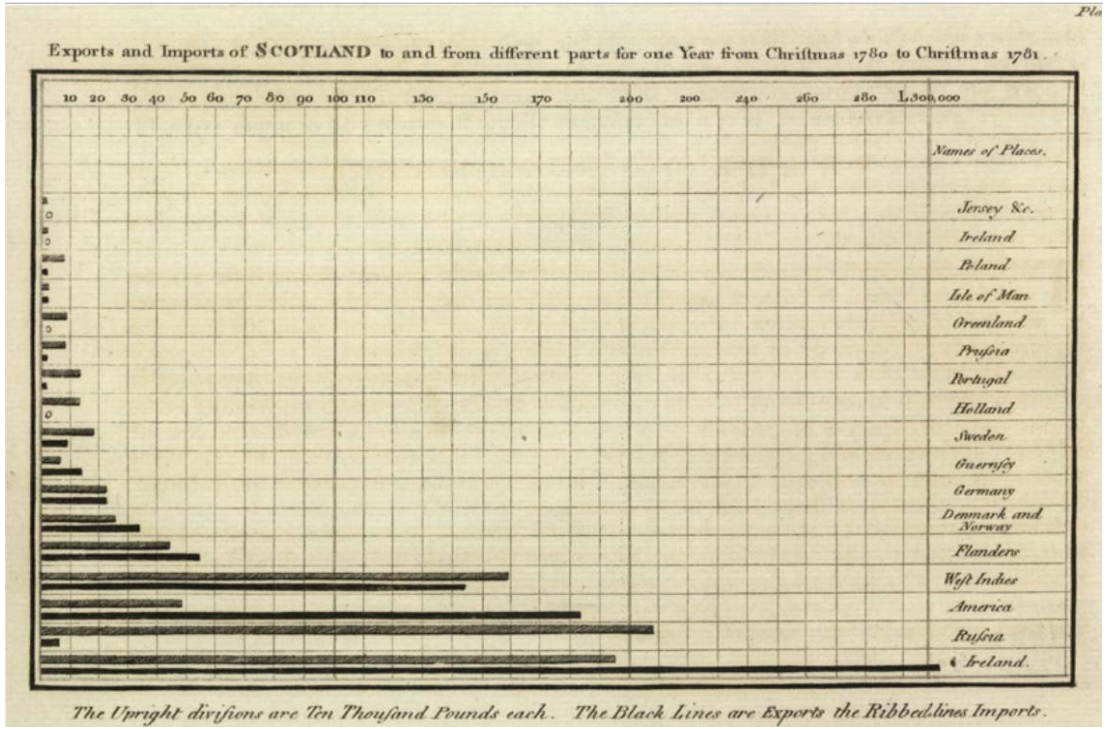
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Bar charts

- Invented by the Scottish engineer and statistician William Playfair (1759–1823)



- also invented
  - line graphs
  - pie charts
    - [Playfair 1786]



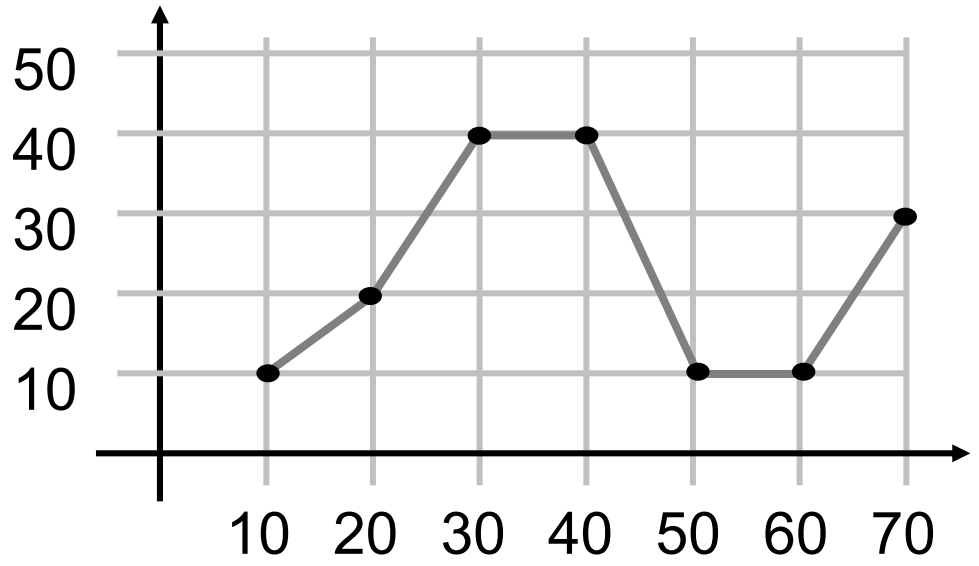
Exports and Imports of Scotland to and from different parts for one year from Christmas 1780 to Christmas 1781

William Playfair 1786

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Bivariate data: line charts

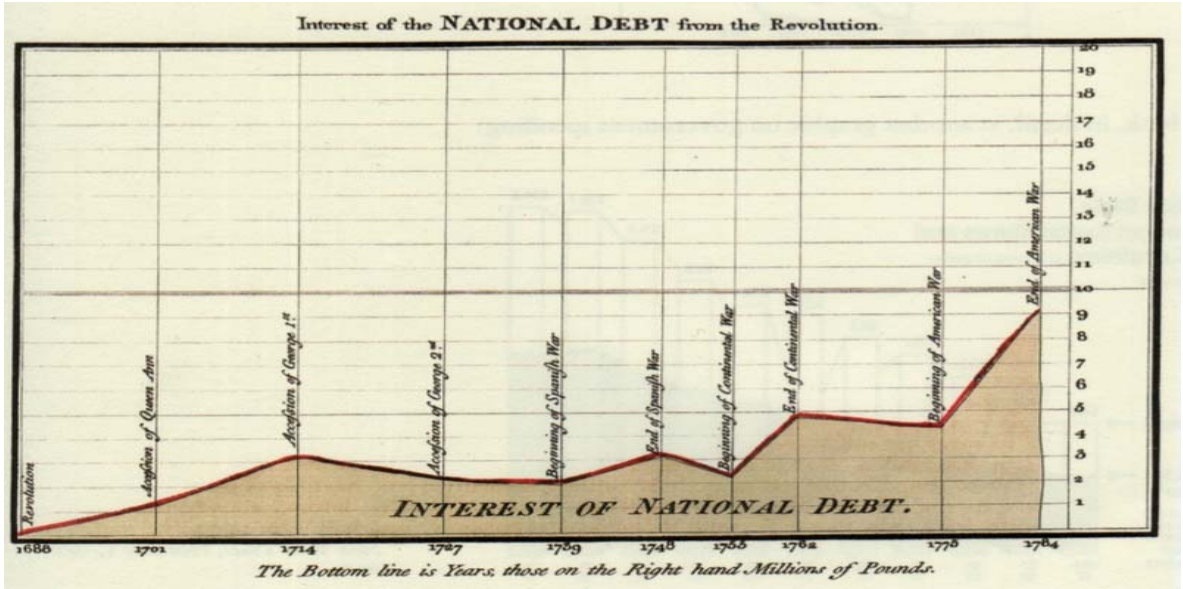
- Line charts or line graphs
  - two quantitative variables
    - one independent and one dependent
    - both continuous
      - the line implies that there are also intermediate values
  - benefits from position but not from length

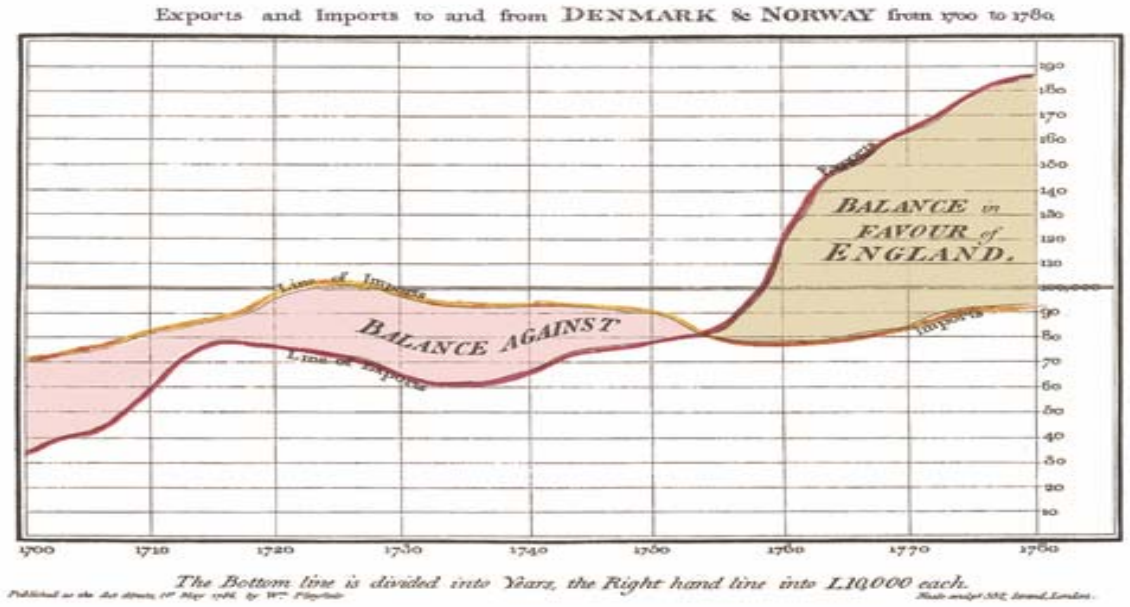


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Line graphs

- Also invented by William Playfair (1759–1823)
  - [Playfair 1786]



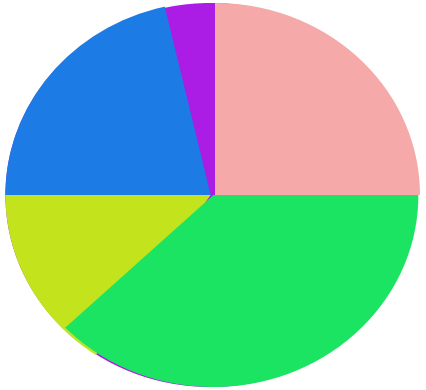


Playfair, “Commercial and Political Atlas”, 1786

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Bivariate data representations

- Pie charts
  - one categorical and one quantitative variable
    - one independent and one dependent
    - benefits from angle and hue
    - shows percentages of the quantitative variable
  - also introduced by William Playfair
    - [Playfair 1801]



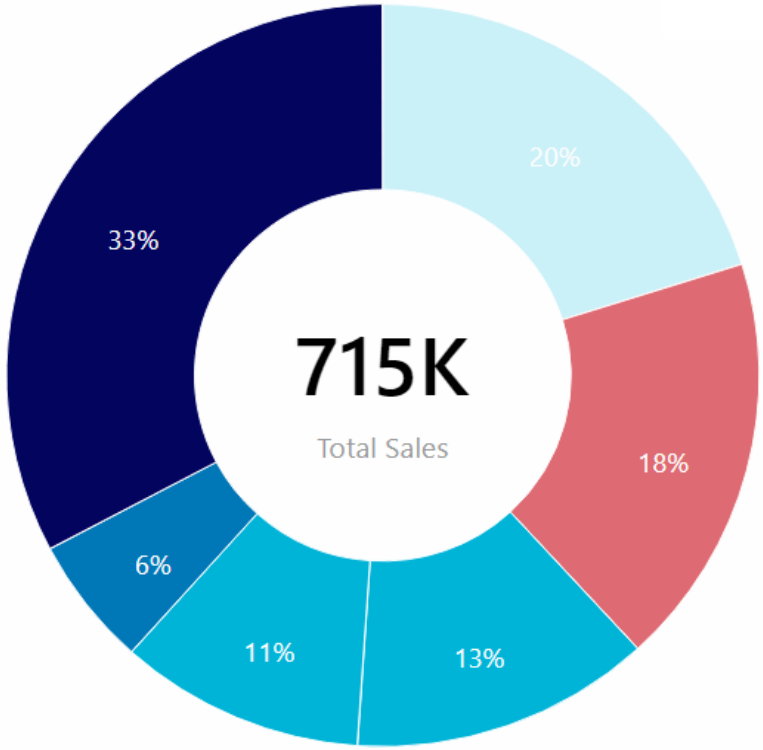


Playfair, “The Statistical Breviary”, 1801

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Bivariate data representations

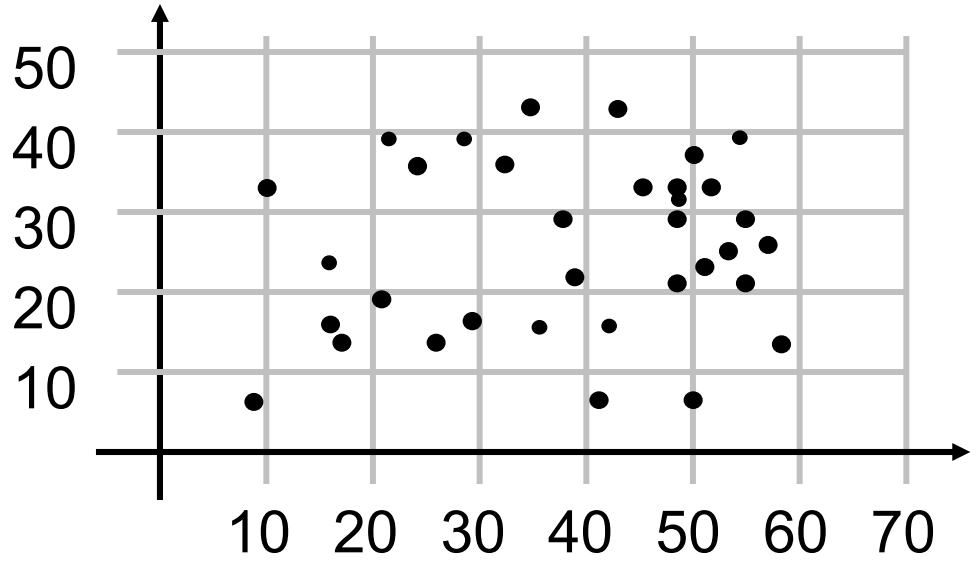
- Donut charts
  - or doughnut charts
  - variant of pie charts
  - help avoid misunderstanding with respect to the areas involved in a pie chart
    - de-emphasizes the use of area

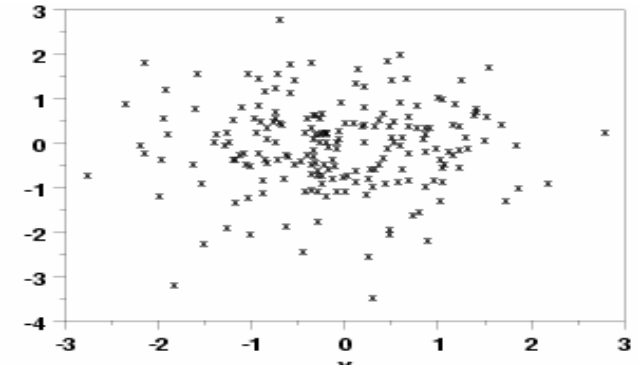


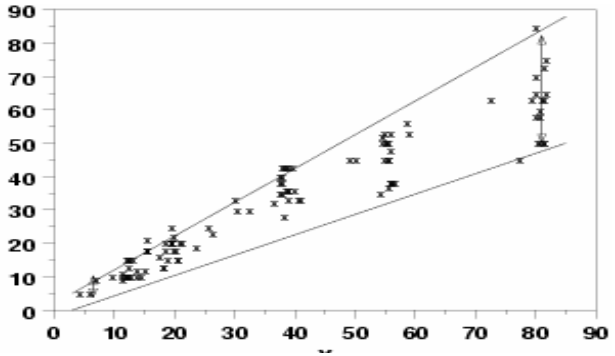
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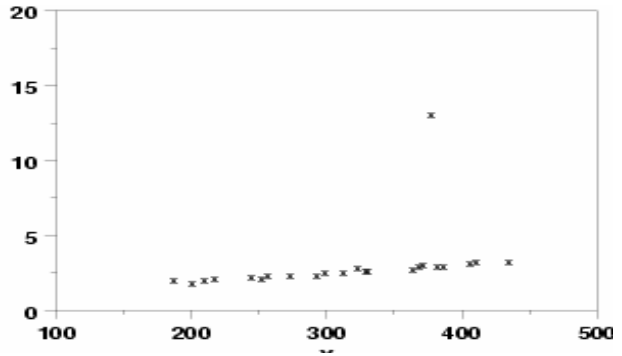
Bivariate data: scatterplots

- Scatterplots or scattergraphs
  - two quantitative variables
    - both independent
    - both continuous
  - mapped to positions
    - but clusters also yield density
  - shows relationship and outliers







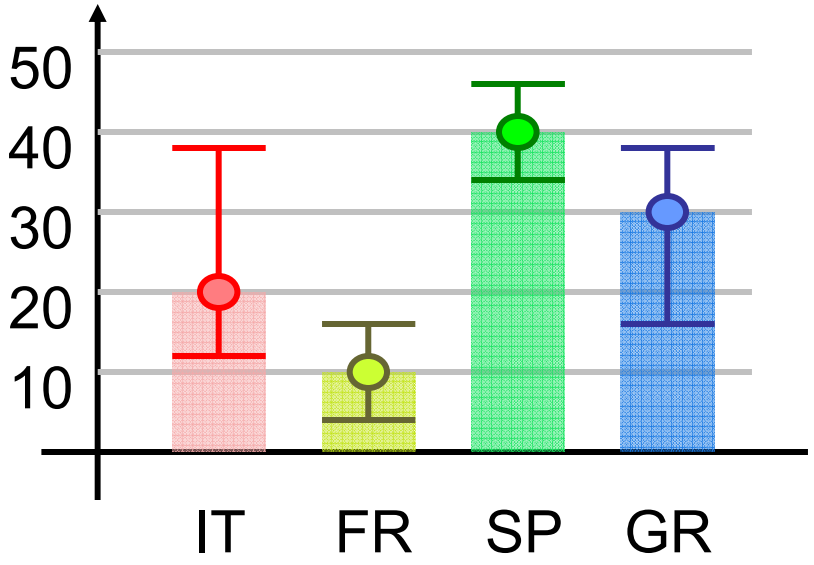


From http://www.itl.nist.gov/div898/handbook/eda/section3/eda33q.htm

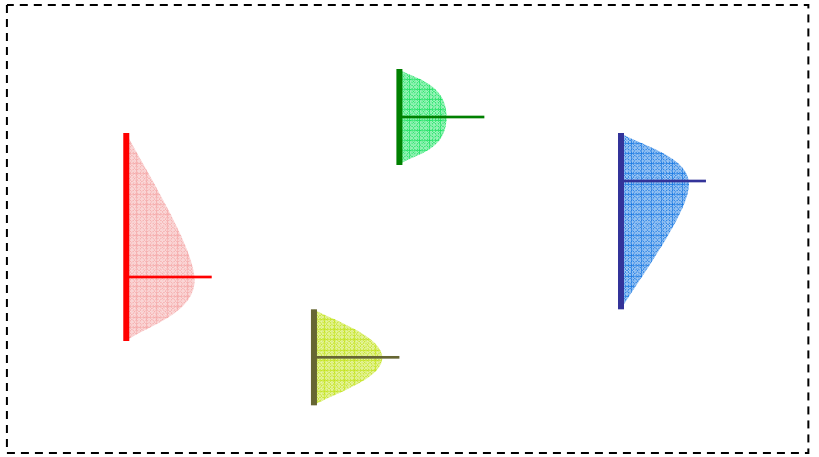
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Bivariate data: bar charts with errors

- Use a bar chart to represent mean or median
- Use glyphs to represent standard deviation or variance



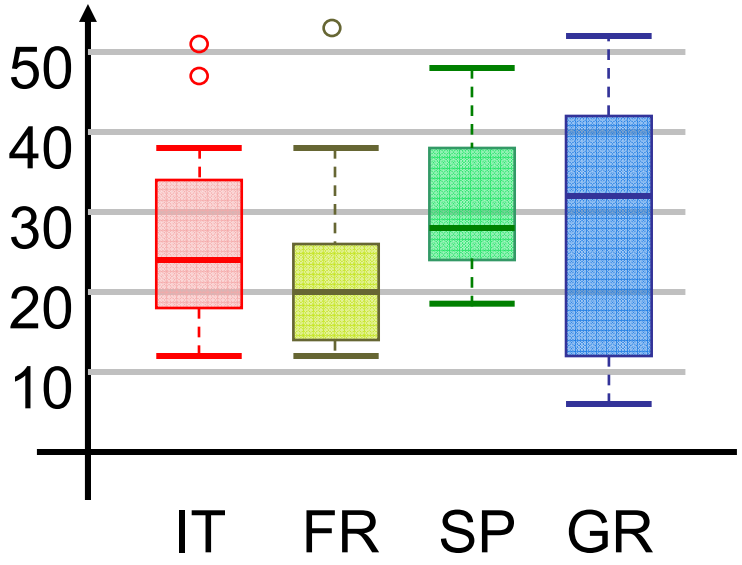
An interpretation of the distributions

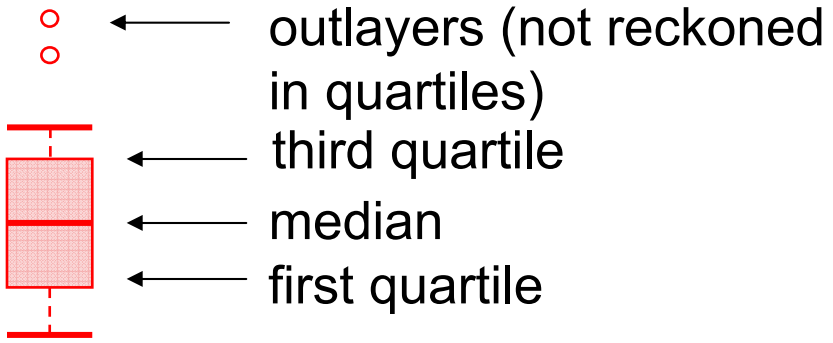


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Box plots

- Also called “box plots with whiskers”
- Represent numerical data through their quartiles
- Several conventions for the upper and lower whiskers
  - minimum and maximum
  - 9th and 91st percentile
  - 2nd and 98th percentile
  - …

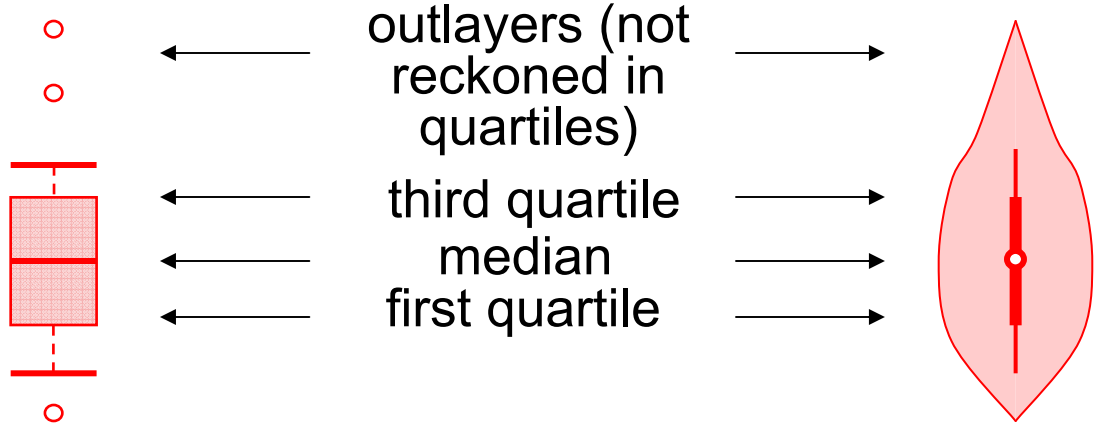


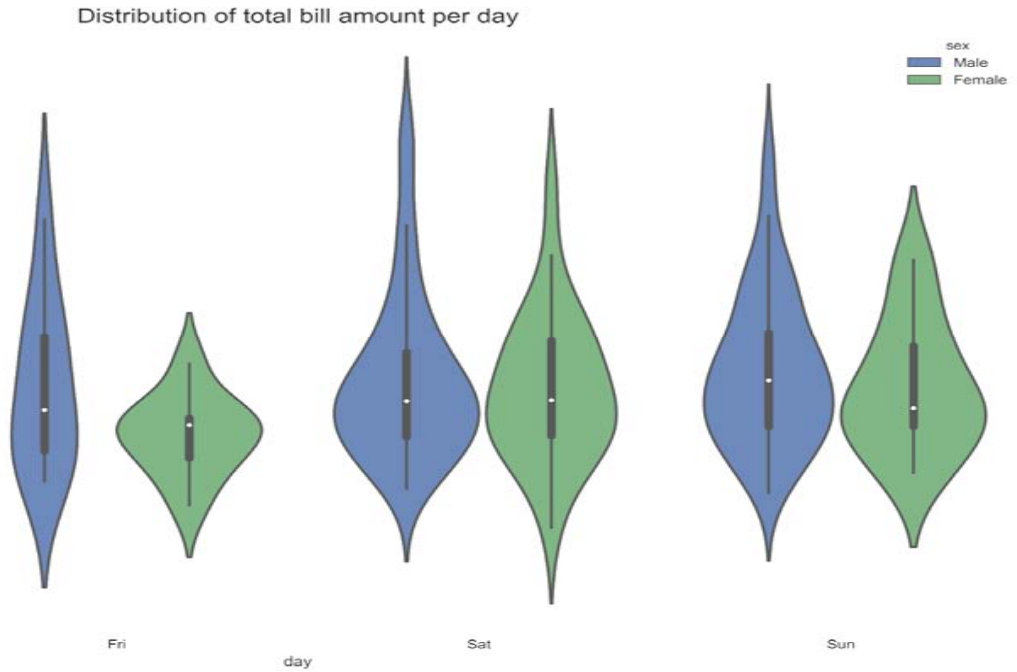


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Violin plots

- Encodes all information of a box plot and also gives the actual distribution



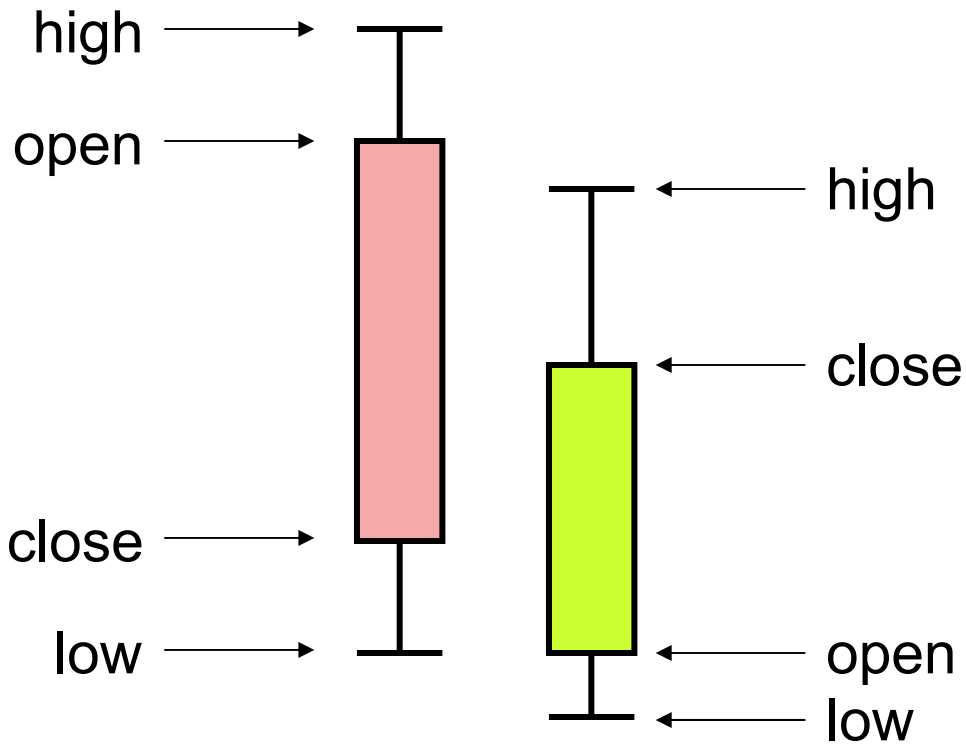


https://towardsdatascience.com/violin-plots-explained-fb1d115e023d

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Candlestick charts

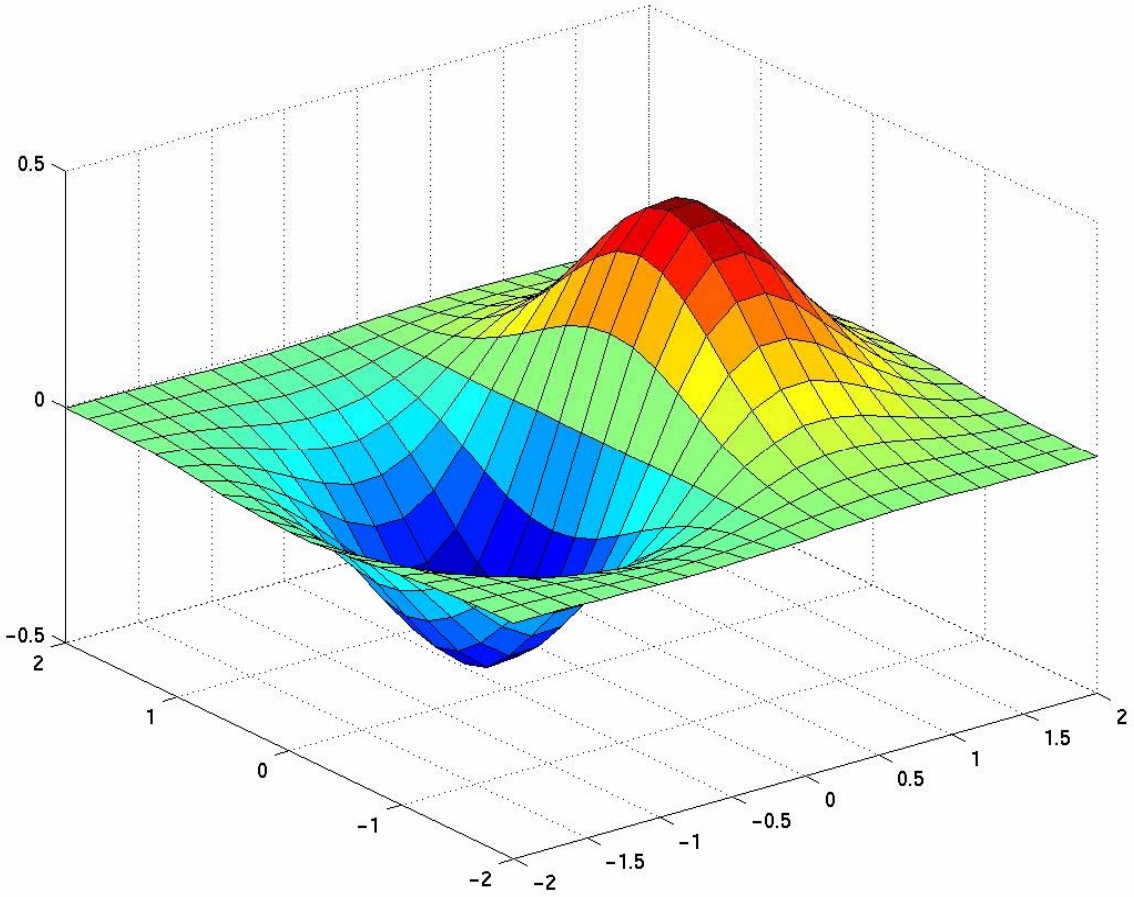
- Also called “Japanese candlestick charts”
- Graphically similar to box plots but with complete different semantics
- Financial charts used to represent the four main prices of a good in the stock exchange



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Trivariate data representation

- 3D (surface) graphs
  - three quantitative variables
    - all three continuous
    - two independent and one dependent
  - benefits from positions
    - the example on the side also uses hue



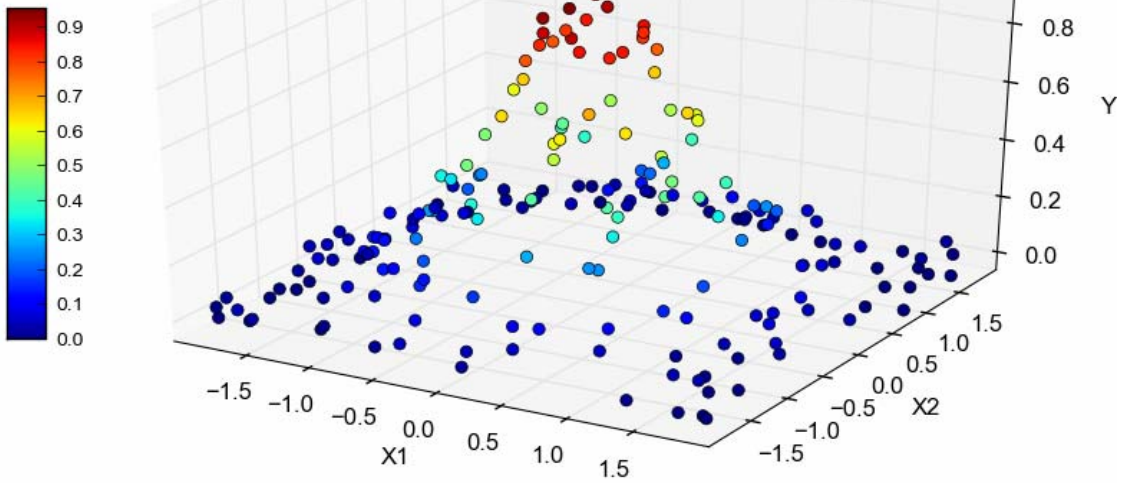
From [Schiff 03]

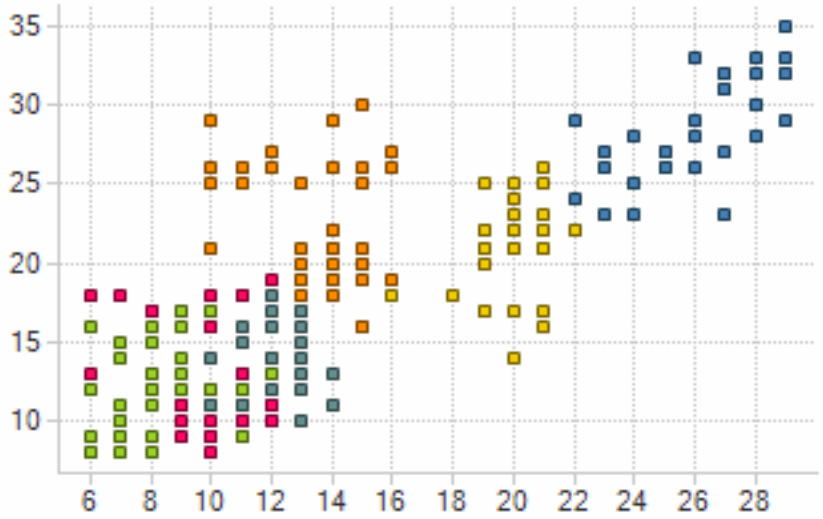
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Scatterplots for trivariate data

- 3D Scatterplots
  - three quantitative variables
- Colored scatterplots
  - two quantitative and a categorical variable
    - example: sales (color is product)

http://www.curveexpert.net/gallery-graphexpert-pro/



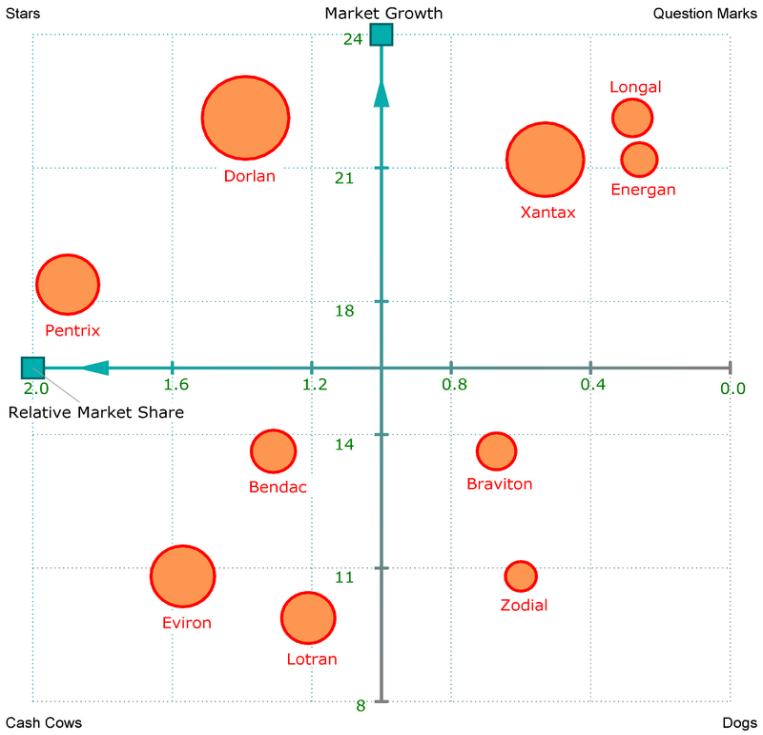


http://custom-analytics.thomsonreuterslifesciences.com/SpotfireWeb/Help/dxpwebclient/scat_what_is_a_scatter_plot.htm

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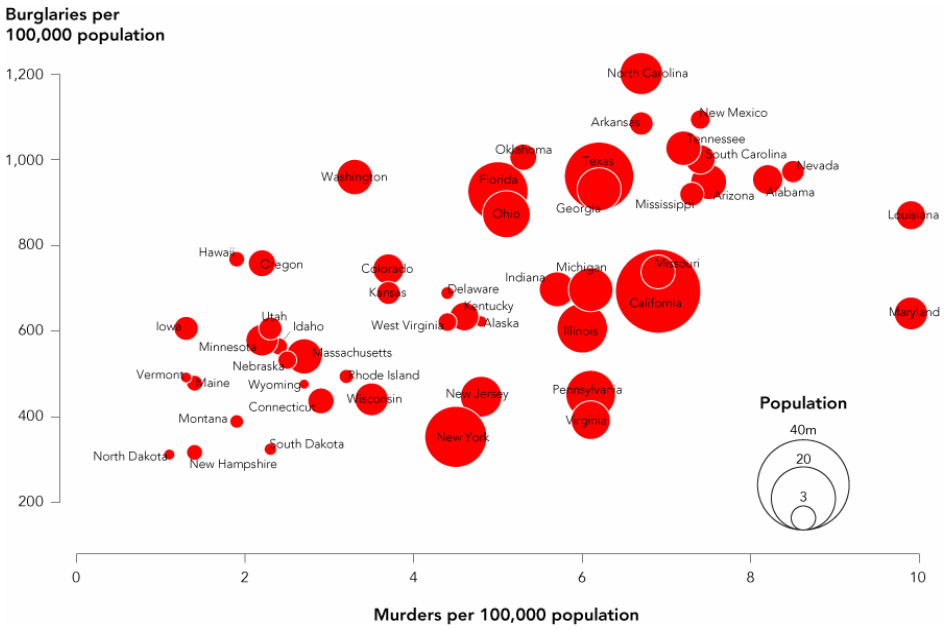
Trivariate data: bubble charts

- Bubble charts
  - three quantitative variables



From Wikipedia

From [Yau '10]



- sometimes called *Boston matrix* from the diagram proposed by the Boston Consulting Group in 1970 to analyze product lines

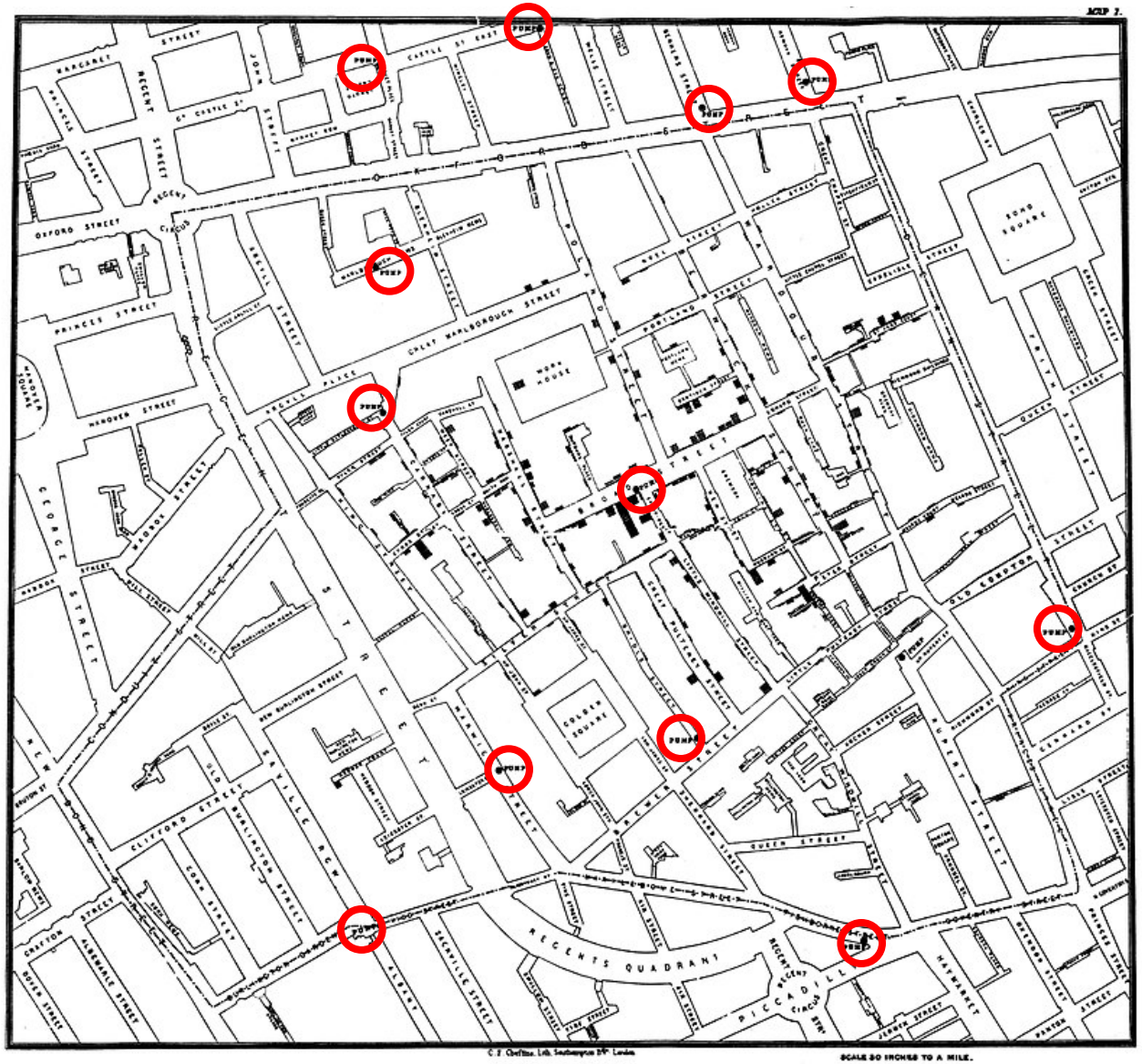
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Spatial datasets visualization

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Thematic maps

- A thematic map is a type of map specifically designed to show a particular theme connected with a specific geographic area



John Snow's cholera map about the cholera deaths in London in the 1840s, published 1854

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Kinds of thematic maps

- Univariate thematic maps
  - non-location data is all of the same kind
    - examples: population density, cancer rates, annual rainfall, etc.
- Bivariate thematic maps
  - mapping shows the geographical distribution of two distinct sets of data
    - example: a map showing both humidity and temperature may be used to explore a possible correlation between the two phenomena
- Multivariate thematic maps
  - non-location data involves more than two sets of data
    - example: a single map showing rainfall, temperature and pressure

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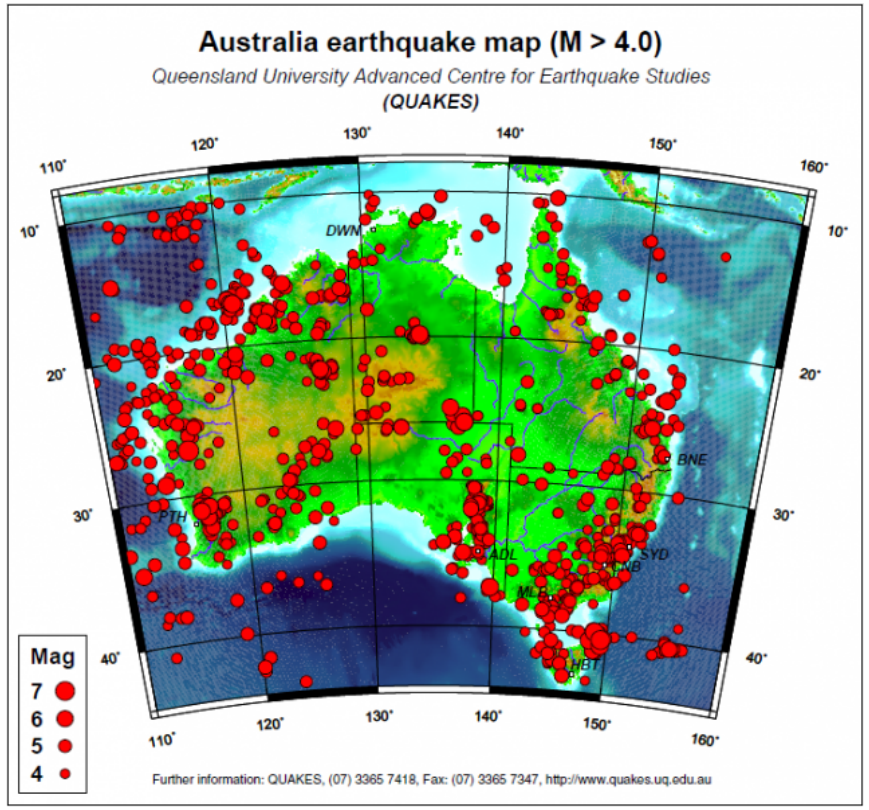
Kinds of thematic maps

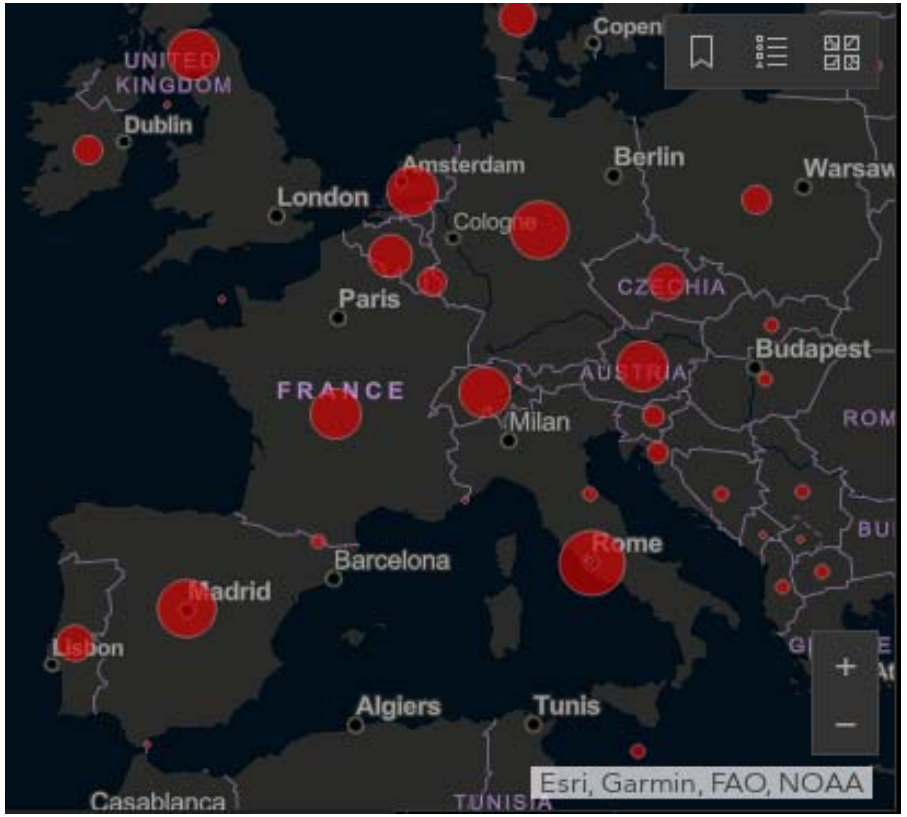
- Point thematic maps
  - non-location data is assigned to points of the map
    - examples: temperature, pressure, annual rainfall, city positions, earthquakes, etc
- Area thematic maps
  - non-location data is assigned to areas of the map
    - examples: population density, cancer rates, internal gross product, etc

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Proportional symbol technique

- Data associated with different areas or locations is represented by symbols of different heights, lengths, areas, or volumes
  - example: earthquakes, city sizes, etc



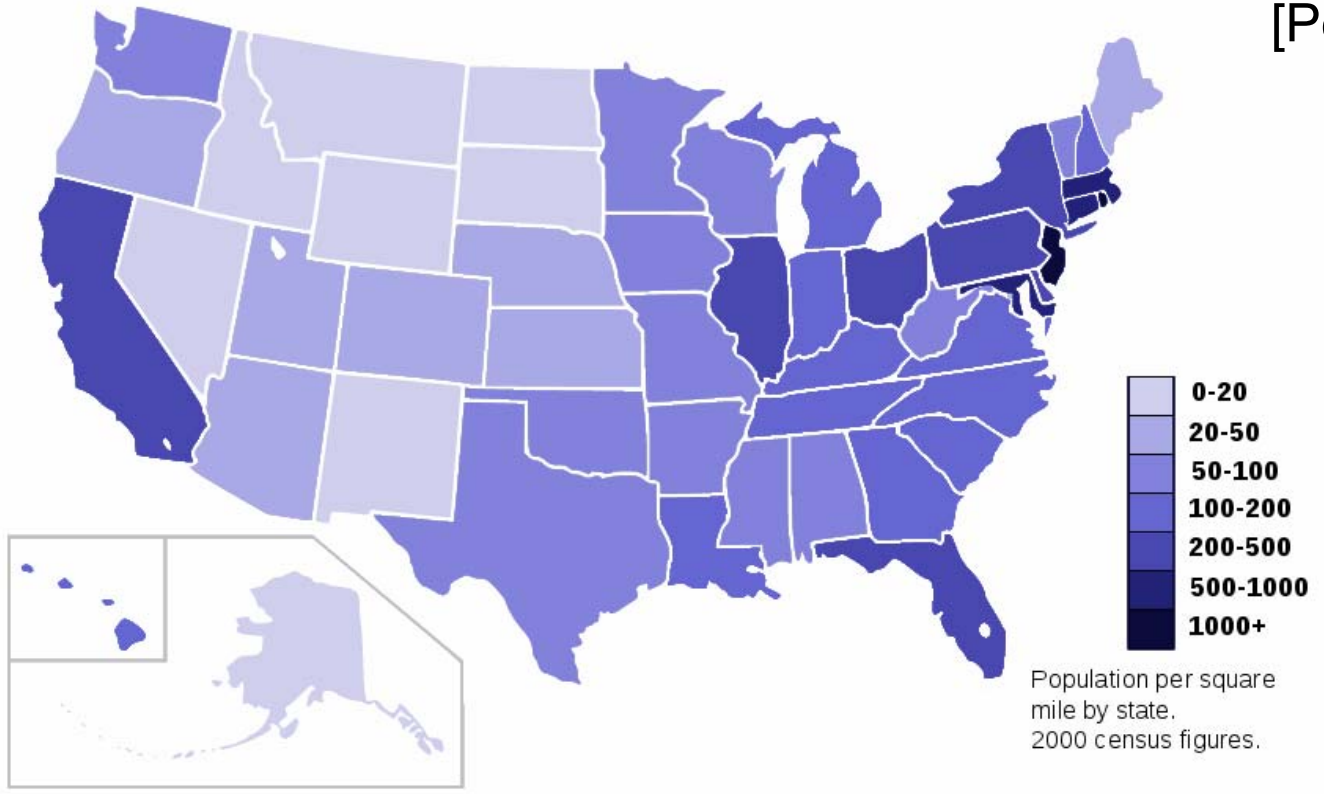


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Choropleth maps

- A *choropleth map* is a thematic map in which areas are shaded or patterned in proportion to a quantitative variable, such as population density or per-capita income

[Peterson 19]

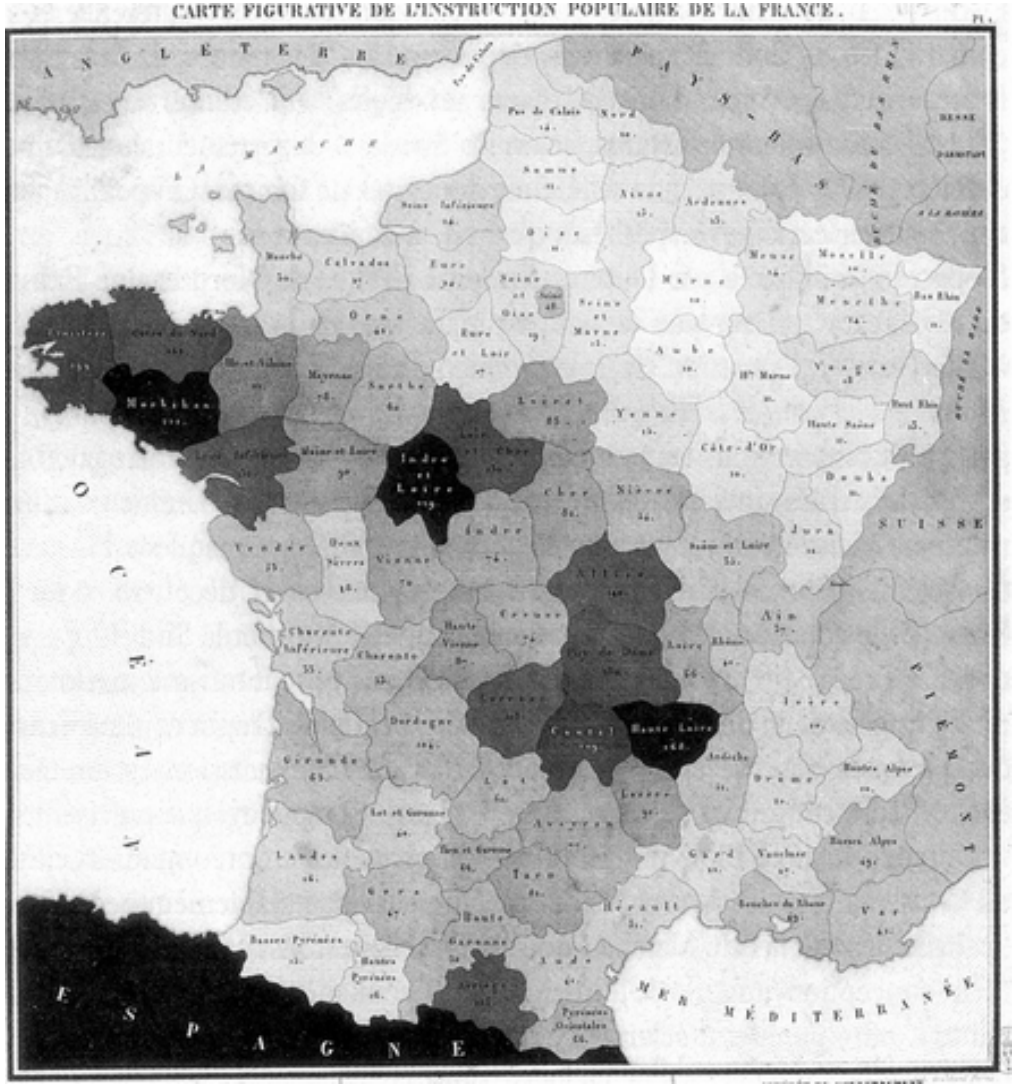


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Choropleth maps history

- Created for the first time in 1826 by Baron Pierre Charles Dupin to show the distribution of illiteracy in France



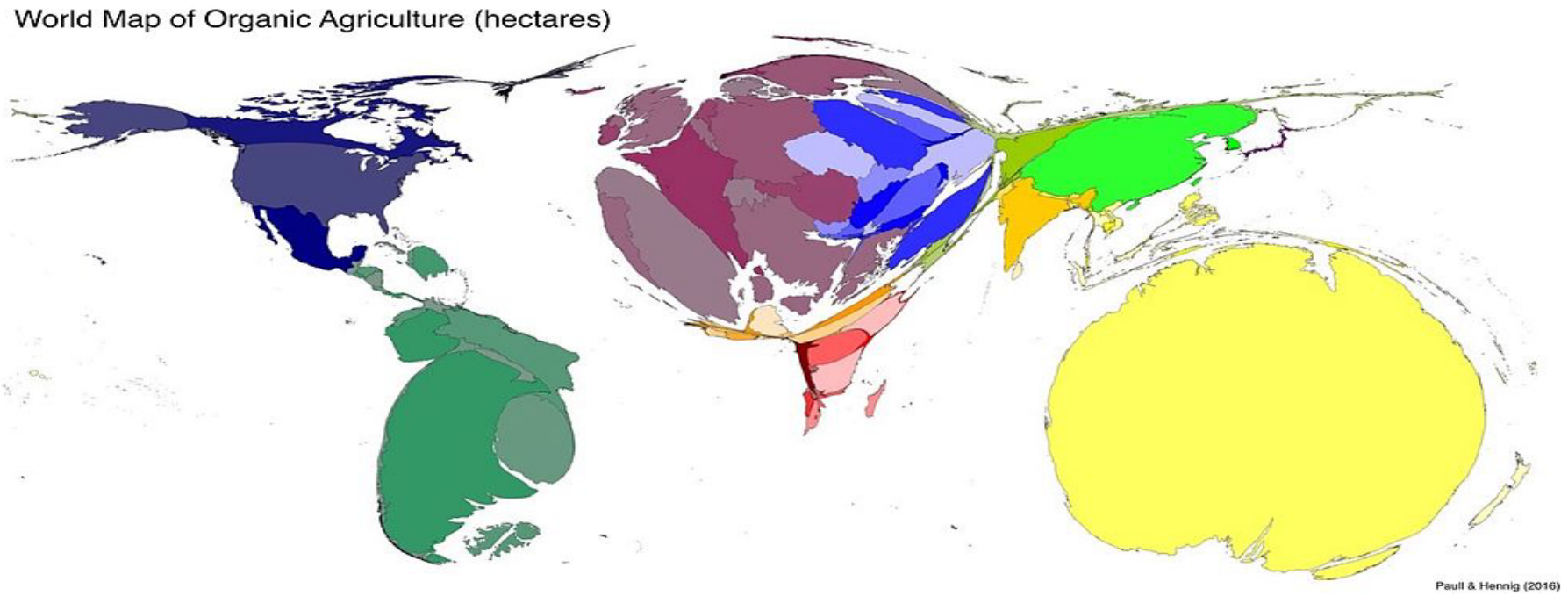


Wikipedia

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Cartograms

- A *cartogram* is a map in which land areas or distances are distorted in order to convey the data associated with locations or areas

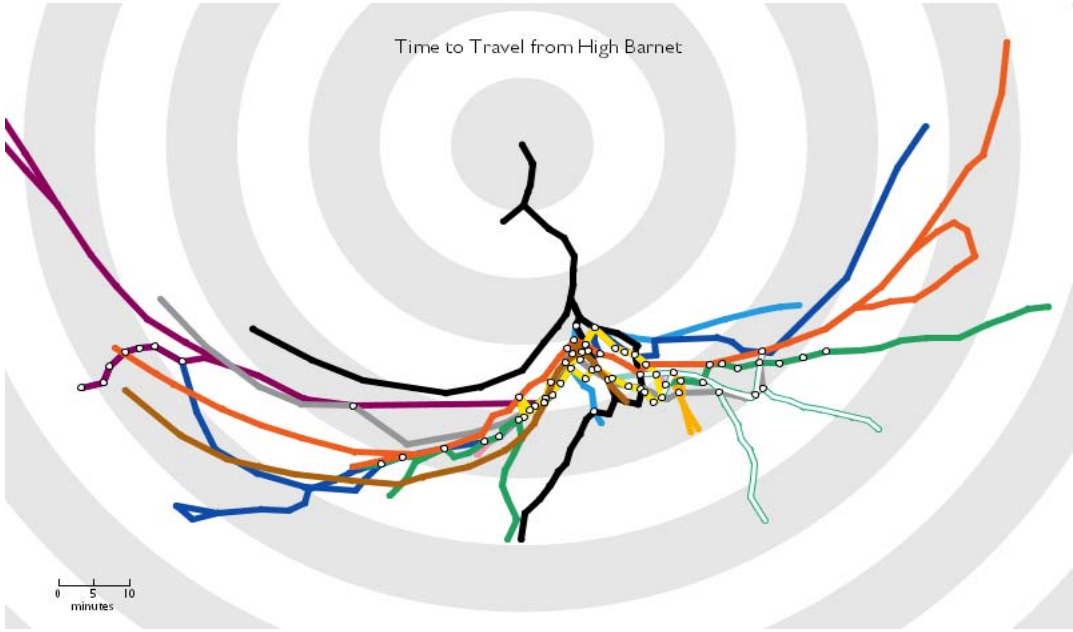


John Paull and Benjamin Hennig - http://orgprints.org/30187/

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Types of cartograms

- Linear cartograms
  - distances are scaled according to the associated values
- Area cartograms
  - areas are scaled according to the associated values



Travel Time Tube Map

http://www.tom-carden.co.uk/p5/tube_map_travel_times/applet/

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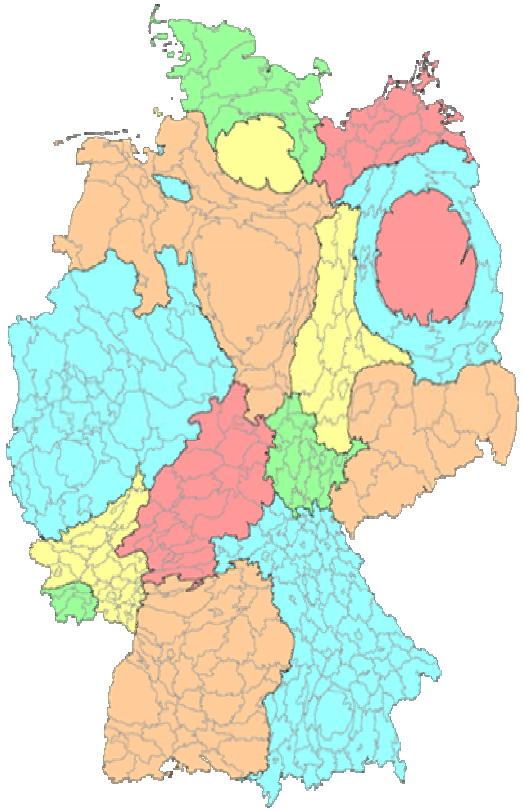
Area cartograms

- Example of area cartogram
  - districts of Germany



Districts of Germany (actual size)

https://germanymap360.com/germany-region-map



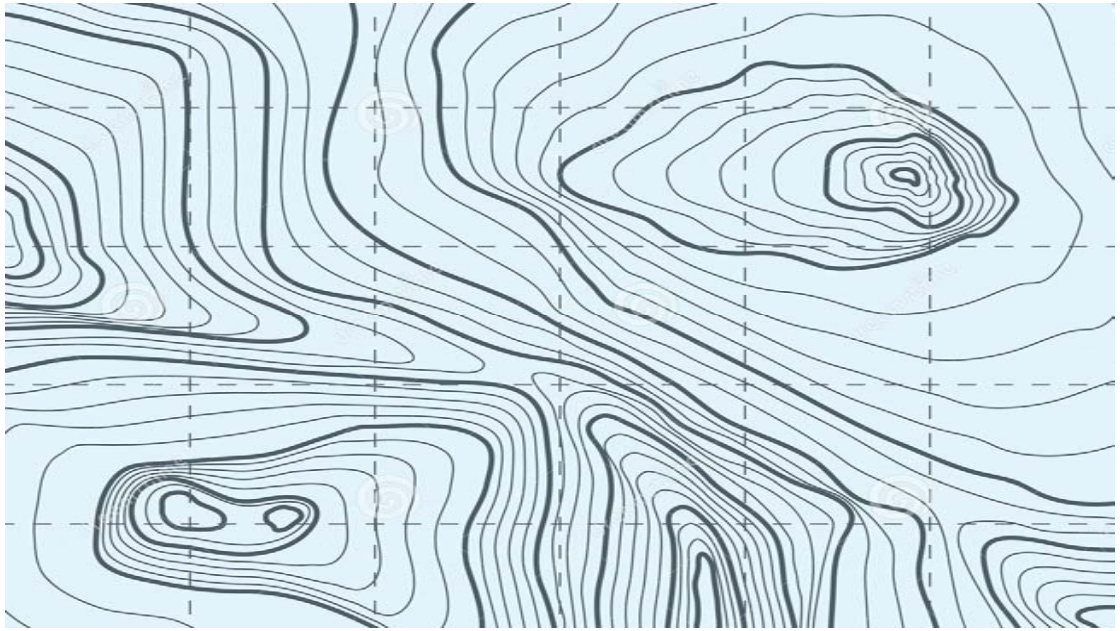
Districts of Germany resized according to population

Christoph Stoepel (wikipedia)

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Contour maps

- Also called *isoline maps* or *isarithmic maps*
- Each contour line
  (also called *isoline*,
  isopleth, *isarithm*)
  is a curve
  representing
  points where a
  quantitative variable
  has a constant value
  - example: elevation maps



https://www.dreamstime.com
image117142736

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Bibliography and further readings

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- [Schiff 03] Jeremy Schiff “Brief Introduction to Matlab”, 2003 http://u.math.biu.ac.il/~schiff/Teaching/276/intmat.html

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